CS 289A Fall 2026

Grad Project Progress

Matthew Bernard, Shiqi Yang *November 6, 2026*

Decisions we made

The Oct 28 meeting settled the project's experimental design. The relevant decisions were:

#	Decision	Options	Our lean
1	Dataset	`derek-thomas/ScienceQA` (all) vs `lmms-lab/ScienceQA-IMG` (image-only)	Image-only for main experiments
2	Softmax variant	One forward pass reading $P(A/B/C/D)$ vs $k = 8$ sampling only	Both; softmax as additional variant
3	Baselines	Oracle (ground-truth labels) + $\tau = 0$ (no filtering)	Both, adopted from Shiqi's suggestion
4	VQAv2 scope	Main / stretch / drop	Stretch (ScienceQA first)

The dataset decision was the only genuinely open item. `lmms-lab/ScienceQA-IMG` was image-only by construction (every row had an image, zero text-only), whereas `derek-thomas/ScienceQA` added the text-only half. Since the project was about a **V**LM, image-only became the main experiment and text-only a secondary analysis. The two datasets were row-for-row aligned on the image+text half, so switching between them was lossless for the vision experiments.

Progress since the meeting

All six action items from the meeting were completed.

#	Action item	Owner	Status
1	Confirmed SCF GPU allocation	Matt	Done — one NVIDIA TITAN Xp (12 GB), driver 570.124.06, CUDA 12.8. Partition `gpu` had `PreemptMode=REQUEUE`, so jobs were checkpointed.
2	Set up Python env on SCF; confirmed SmolVLM loads	Matt	Done — conda env `smolvlm` (Python 3.11), torch 2.14.1+cu126, transformers 5.x. SmolVLM-500M loaded at `cuda:0`, `torch.bfloat16`, 507.48M params.
3	HF Hub repo for checkpoints	Matt	Done — `mattbix/cs289-smolvlm-checkpoints`, with both baseline JSONLs under `baselines/`.
4	`scripts/01_baseline_eval.py` — zero-shot greedy on 500-question subset	Matt	Done — results are reported below.
5	ScienceQA counts, image+text vs. text-only, per split	Matt, Shiqi	Done — counts are reported below; Matt ran the counts, Shiqi reviewed.
6	SmolVLM forward-pass API notes for softmax over options	Matt, Shiqi	Done — `scripts/02_softmax_eval.py`; Matt implemented and ran it, Shiqi reviewed.

Two notes on the infrastructure. First, the `gpu` partition's single TITAN Xp was small but sufficient: SmolVLM-500M occupied about 1.1 GB at bf16 inference, so LoRA fine-tuning fit comfortably. The real constraint was `PreemptMode=REQUEUE` — long runs were checkpointed per example, which the baseline scripts did (JSONL append after every question). Second, the HF Hub token was stored under `$HOME` (`~/.cache/huggingface/token`), so it survived login-node changes; `HF_HOME` was exported from both `~/.bashrc` and `~/.profile` to cover interactive `srun` and `sbatch` contexts alike.

Preliminary results

Dataset counts

Dataset	Split	Total	Image+text	Text-only
`lmms-lab/ScienceQA-IMG`	train	6,218	6,218	0
	validation	2,097	2,097	0
	test	2,017	2,017	0
`derek-thomas/ScienceQA`	train	12,726	6,218	6,508
	validation	4,241	2,097	2,144
	test	4,241	2,017	2,224

The image+text column of `derek-thomas/ScienceQA` matched `lmms-lab/ScienceQA-IMG` row-for-row, confirming the two datasets were aligned. The main experiments therefore used `lmms-lab/ScienceQA-IMG`; the text-only rows were held out for a secondary analysis.

Baselines (500-question image-only test subset, seed 0)

Both baselines were evaluated on the same 500-question random subset of `lmms-lab/ScienceQA-IMG` test, so row-by-row comparison was valid.

Variant	Accuracy	Forward passes per question
Zero-shot greedy	0.6480 (324/500)	1 autoregressive decode, ≤ 8 tokens
Zero-shot softmax argmax	**0.6700** (335/500)	1 single pass, no decode

The softmax variant was both cheaper (one forward pass versus an autoregressive decode) and slightly more accurate (+2.2 points, 11 more correct). That was the headline result of the ablation: the graded-confidence signal captured information the sampled-mode signal did not.

Ablation split (greedy vs. softmax, same 500)

Outcome	Count
Agreed on predicted letter	373 / 500 (74.6%)
Both right	276
Greedy-only right	48
Softmax-only right	59

Check: $276 + 48 = 324$ (greedy total), $276 + 59 = 335$ (softmax total).

The two methods disagreed on a quarter of the questions (127/500), and on those disagreements softmax was right slightly more often (59 vs 48). Neither method dominated. This supported running both as an ablation rather than choosing one: the sampling-based consistency signal and the softmax-confidence signal were complementary, not redundant. The 107 questions on which exactly one method was correct were the recoverable headroom for the consistency-filter fine-tuning stage.

Remaining work

The meeting's action items were closed before the progress report was due. The remaining research milestones would proceed as follows.

1. **Consistency-filter fine-tuning.** The filter at threshold τ would be implemented, and the two comparison baselines (oracle on ground-truth labels; $\tau = 0$, no filtering) agreed at the meeting would be run. This was the experiment that would isolate the contribution of the filter.
2. **SmolVLM pseudo-label generation.** The $k = 8$ sampled outputs on the training split would be generated, with mode and frequency extracted, and the high-consensus subset selected.
3. **LoRA fine-tuning.** LoRA on SmolVLM-500M at bf16 would be run; the TITAN Xp's 12 GB would be sufficient.
4. **Post-fine-tuning evaluation.** Both baselines would be re-run on the fine-tuned model and compared against the 0.6480 / 0.6700 figures above.
5. **Write-up.** The two baseline tables and the ablation split above would become the paper's first results section; the consistency-filter comparison would become the second.

Collaboration

Repo: github.com/mattbix/cs289gradproject. Checkpoints: huggingface.co/mattbix/cs289-smolvlm-checkpoints (both baseline JSONLs were uploaded under `baselines/`). The weekly check-in cadence was as agreed at the meeting.